



// INSTALLATION STEPS

Ubuntu 22.04 + ROS 2 Humble + Gazebo Fortress

Everything to install before the workshop, from a fresh OS to a working simulation, with a check at the end to confirm it all runs.

```
user@robot:~$ ros2 run demo_nodes_cpp talker
```

OPERATING SYSTEM

Ubuntu 22.04

Jammy Jellyfish · native install

MIDDLEWARE

ROS 2 Humble

Humble Hawksbill · LTS

SIMULATOR

Gazebo Fortress

installed via ros_gz

00

Before you begin

IMPORTANT

We strongly recommend that workshop attendees do **not** use a Virtual Machine for this setup. Install Ubuntu natively, either as a dual boot or on a dedicated drive.

- ☐ Install **Ubuntu 22.04 LTS** (Jammy Jellyfish).
- ☐ **ROS 2 Humble Hawksbill** is built for Ubuntu 22.04 (Jammy). Other Ubuntu versions will not work with these steps.
- ☐ Back up your data before you partition a disk for dual boot.

01

Install Ubuntu 22.04

Download the desktop image, write it to a USB drive, and follow the official installer walkthrough.

DOWNLOAD Ubuntu 22.04 Desktop
<https://releases.ubuntu.com/22.04/>

GUIDE Installation guide
<https://ubuntu.com/tutorials/install-ubuntu-desktop>

02

Install ROS 2 Humble

Follow the official Debian-package guide. It adds the ROS apt repository and installs `ros-humble-desktop`.

DOCS ROS 2 Humble • Ubuntu (deb)
<https://docs.ros.org/en/humble/Installation/Ubuntu-Install-Debs.html>

03 Environment setup

Once ROS 2 Humble is installed, set up your environment by sourcing its setup file.

SOURCE ROS IN THE CURRENT TERMINAL

```
● ● ● bash

$ source /opt/ros/humble/setup.bash
```

LOAD IT AUTOMATICALLY IN EVERY NEW TERMINAL

```
● ● ● bash

$ echo "source /opt/ros/humble/setup.bash" >> ~/.bashrc
$ source ~/.bashrc
```

NOTE

Running `source ~/.bashrc` reloads the updated file. Closing all terminals and opening a new one does the same thing.

04 Gazebo Fortress

Fortress is the Gazebo release paired with ROS 2 Humble. You don't need a separate install:

`ros-humble-ros-gz` in step 05 installs it for you. On Fortress the command is `ign gazebo`.

DOCS Fortress · ROS integration

https://gazebosim.org/docs/fortress/ros_installation/

DOCS Getting started with Fortress

<https://gazebosim.org/docs/fortress/getstarted/>

05 ROS ↔ Gazebo integration

Install the pairing packages that let ROS 2 nodes talk to Gazebo simulations. Reference: gazebosim.org/docs/fortress/ros_installation

DEFAULT ROS-GAZEBO PAIRING

```
● ● ● bash

$ sudo apt-get install ros-humble-ros-gz
```

BRIDGE FOR TOPIC COMMUNICATION BETWEEN ROS 2 AND GAZEBO

```
● ● ● bash

$ sudo apt-get install ros-humble-ros-gz-bridge
```

ROS2_CONTROL PLUGIN FOR LOADING AND RUNNING CONTROLLERS IN GAZEBO

```
● ● ● bash

$ sudo apt-get install ros-humble-ign-ros2-control ros-humble-ign-ros2-control-demos
```

06 Key packages

The packages used throughout the workshop for control, robot description, navigation and mapping.

```
● ● ● bash

# ros2_control (controller manager, controllers)
$ sudo apt install ros-humble-ros2-control ros-humble-ros2-controllers
# xacro (robot description macros)
$ sudo apt install ros-humble-xacro
# Navigation2 (Nav2) + bringup
$ sudo apt install ros-humble-navigation2 ros-humble-nav2-bringup
# SLAM
$ sudo apt install ros-humble-slam-toolbox
# Twist mux and Twist stamper
$ sudo apt install ros-humble-twist-mux
$ sudo apt install ros-humble-twist-stamper
# OpenCV + cv_bridge, keyboard teleop
$ sudo apt install ros-humble-cv-bridge python3-opencv ros-humble-teleop-twist-keyboard
```

07 Verify the installation

After finishing all the steps above, run these in **new** terminals to confirm that ROS and Gazebo both work.

TERMINAL A • C++ TALKER

```
terminal A

$ ros2 run demo_nodes_cpp talker
[INFO] [talker]: Publishing: 'Hello World: 1'
```

TERMINAL B • PYTHON LISTENER

```
terminal B

$ ros2 run demo_nodes_py listener
[INFO] [listener]: I heard: [Hello World: 1]
```

The talker should report that it is publishing messages, and the listener should report that it heard them. This confirms that both the C++ and Python APIs work.

GAZEBO FORTRESS

```
bash

$ ign gazebo shapes.sdf
```

This starts the Gazebo server and GUI with a sample world containing three simple shapes.



Quick fixes

SYMPTOM	TRY THIS
ros2: command not found	The terminal has not sourced ROS. Run <code>source /opt/ros/humble/setup.bash</code> or redo step 03.
Unable to locate package ros-humble-...	The ROS apt repository was not added, or you are not on Ubuntu 22.04. Recheck step 02 and run <code>sudo apt update</code> .
gz: command not found	Fortress uses the <code>ign</code> command: run <code>ign gazebo</code> , not <code>gz sim</code> .
Listener hears nothing	Make sure both terminals are sourced and have the same <code>ROS_DOMAIN_ID</code> (check with <code>printenv ROS_DOMAIN_ID</code>).



You're all set.

If all three checks pass, ROS 2 Humble and Gazebo Fortress are installed and working on your system.